
How Representation Curvature Organizes Photometric Decoding in Astronomical Foundation Models

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Abstract

1 Astronomical foundation models are routinely read out with linear probes, yet it
2 remains unclear how scientifically useful information is organized in their repre-
3 sentation geometry. We study a frozen ViT-B galaxy embedding and a photometric
4 r -band magnitude target, constructing local quadratic charts on the unit sphere and
5 a split-half sphere-normal mean-curvature statistic K_H^{cross} . At a frozen chart rank
6 and neighbourhood bandwidth ($d=16$, $k=2048$, $n=512$ anchors), greater curva-
7 ture energy is associated with worse global linear decoding (controlled Spearman
8 $\rho \approx -0.240$ for local OOF R^2 and $\rho \approx +0.227$ for MSE). The same neighbour-
9 hoods exhibit held-out quadratic structure in chart coordinates: an unrestricted
10 quadratic label map improves over a tangent-linear map (median $\Delta_Q \approx +0.021$),
11 and that gain increases with curvature ($\rho \approx +0.111$). The label Hessian aligns
12 with high-energy sphere-normal bending modes, but the associated quadratic fam-
13 ily is algebraically full rank: sphere-normal geometry supplies an informative
14 anisotropic prior rather than a low-dimensional constraint. The association is rank-
15 and bandwidth-conditioned, not claimed to be scale-invariant.

16 1 Introduction

17 Linear probes are a standard way to ask whether a foundation-model representation contains a
18 scientific observable [Alain and Bengio, 2017]. In astronomy this question is now asked of large
19 vision embeddings of galaxies [UniverseTBD et al., 2025], against a background hypothesis that
20 foundation models converge toward shared geometric structure [Huh et al., 2024]. Linear decodability,
21 however, need not be uniform across representation space: local extrinsic bending of the activation
22 chart may mark either lost information or information that is organized nonlinearly.

23 We separate those possibilities at a frozen configuration of one ViT-B [Dosovitskiy et al., 2021]
24 galaxy embedding. First we test whether sphere-normal mean-curvature energy co-varies with the
25 error of a *global* linear probe evaluated locally. Then we test whether the photometric label itself
26 has held-out quadratic structure in the same chart coordinates, whether that structure is more useful
27 in high-curvature neighbourhoods, and whether the fitted label Hessian is oriented along the chart’s
28 high-energy bending modes.

29 **Contributions.** (i) At frozen rank and bandwidth, reliable sphere-normal curvature is associated
30 with direct global-probe error, not merely an R^2 denominator effect. (ii) Local photometric labels
31 contain curvature-linked held-out quadratic structure in chart coordinates. (iii) Label Hessians align
32 with sphere-normal bending modes, so geometry acts as an informative quadratic prior rather than an
33 algebraically restricted function class.

34 2 Method

35 **Data, model, and target.** We use a frozen ViT-B embedding of galaxy images (unit-sphere
36 activations) and $n=512$ hash-stable anchors with $k=2048$ neighbours, aligned throughout by object
37 `sample_id`. The scientific target is an astronomical photometric observable—apparent r -band
38 magnitude—not a fundamental physical property and not a spectroscopic DESI label. A five-fold
39 ridge probe ($\alpha=100$) maps embeddings to this magnitude; all probe metrics below are out-of-fold
40 (OOF). The primary global-decoding outcome is the local OOF coefficient of determination R_G^2 of
41 that probe, together with local OOF MSE. Catalog magnitude as a raw vector is never substituted for
42 either quantity.

43 **Charts and curvature.** At an anchor x_0 we form tangent coordinates u in a nested PCA frame J
44 of dimension d after removing the radial component on the unit sphere, following the local-quadratic
45 charting tradition [Donoho and Grimes, 2003]. A regularized quadratic chart

$$f(u) = x_0 + Ju + \frac{1}{2}Q(u, u)$$

46 is fit with split-half cross-evaluation. Tangential warp and the forced unit-sphere radial component
47 are removed, leaving a sphere-normal second-fundamental-form tensor B^S . The sphere-normal
48 mean-curvature vector is $H^S = d^{-1} \sum_a B_{aa}^S$; the production statistic is the split-half inner product
49 $K_H^{\text{cross}} = \langle H^{(A)}, H^{(B)} \rangle$. The cross product avoids the positive noise bias of $\|\hat{H}\|^2$. We call this
50 finite-scale *sphere-normal extrinsic bending*, not intrinsic Riemannian curvature of an unknown
51 manifold. The frozen scientific configuration is $d=16$, $k=2048$.

52 **Quadratic label models.** On the same neighbourhoods and outer OOF folds we compare a tangent-
53 linear map $\hat{y}_L(u) = a_0 + a_1^\top u$ with an unrestricted quadratic $\hat{y}_{UQ}(u) = a_0 + a_1^\top u + \frac{1}{2}u^\top \Gamma u$ ($\Gamma = \Gamma^\top$;
54 $q=136$ coefficients at $d=16$). A sphere-normal-constrained map uses $\gamma_{BS} = B_{\text{flat}}^{S\top} c$. Linear and
55 quadratic blocks are ridge-penalized with nested cross-validation inside each outer training fold,
56 an unpenalized intercept, and a single train-only RMS scale for u . The label Hessian Γ describes
57 second-order variation of the photometric target in local chart coordinates.

58 **Inference.** Controls are log k NN radius, local label variance, and evaluation count. Primary
59 tests use $B=10^4$ rank-space Freedman–Lane permutations and $B=2000$ bootstrap intervals; $p=0$ is
60 reported as $p < 1/(B+1)$. The two primary quadratic hypotheses are Holm-corrected.

61 3 Results

62 **Global linear decoding.** At the frozen configuration, greater K_H^{cross} is associated with worse
63 local OOF linear decoding of photometric magnitude, with $\rho_{\text{ctl}}(K_H^{\text{cross}}, R_G^2) \approx -0.240$ and
64 $\rho_{\text{ctl}}(K_H^{\text{cross}}, \text{MSE}_G) \approx +0.227$ (Fig. 1; $n=512$). The MSE association rules out an R^2 -denominator
65 explanation: curvature tracks higher squared error, not merely a change in local target variance
66 (controlled ρ with local SST ≈ -0.025). The result is conditioned on chart rank and neighbourhood
67 bandwidth (at the same k , controlled R^2 is $+0.143$ at $d=12$ and -0.233 at $d=20$); we do not claim
68 scale invariance.

69 **Held-out quadratic structure.** Define $\Delta_Q = \text{MSE}_L - \text{MSE}_{UQ}$. Median $\Delta_Q \approx +0.021$ (bootstrap
70 95% CI $[0.0197, 0.0216]$); the gain is positive at all 512 anchors ($p_{\text{MC}} < 1/2001$). Curvature
71 predicts that gain: $\rho_{\text{ctl}}(K_H^{\text{cross}}, \Delta_Q) \approx +0.111$ ($p_{\text{MC}} \approx 0.0075$). Both primary tests pass Holm
72 correction (Fig. 2a,b). Local labels therefore exhibit reproducible held-out quadratic structure in chart
73 coordinates, and that structure becomes more predictively useful in neighbourhoods with greater
74 sphere-normal curvature energy. No causal claim is made.

75 **Geometry as an anisotropic prior.** B_{flat}^S is algebraically full rank 136 at every anchor. Median
76 energy ranks are $r_{90}=71$, $r_{95}=90$, $r_{99}=119$; the original constrained model retained 48 modes
77 because of a hard cap, not because 48 modes contain nearly all bending energy. Full-rank ridge on c
78 is equivalent to unrestricted quadratic regression with the geometry-derived penalty $\gamma^\top (B^{S\top} B^S)^+ \gamma$.
79 Sphere-normal geometry therefore supplies an informative anisotropic prior over local label Hessians,
80 not a genuinely low-dimensional quadratic function class. Geometry-only truncations, selected

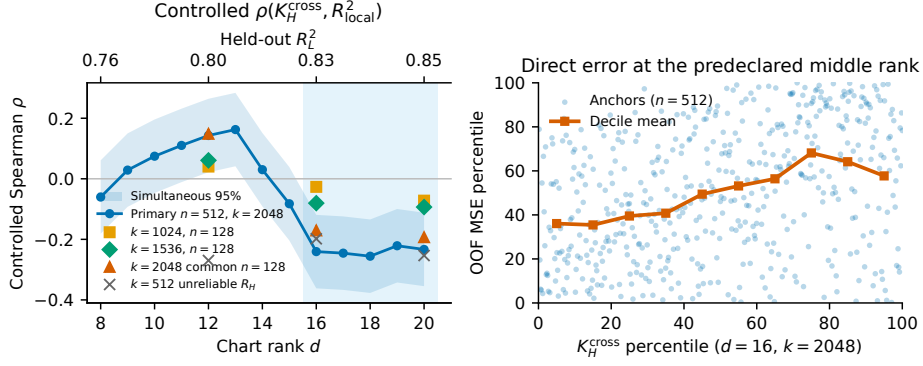


Figure 1: Confirmatory frozen analysis ($n=512, k=2048$). Left: controlled $\rho(K_H^{\text{cross}}, R_G^2)$ versus chart rank; the scientific configuration is the predeclared middle rank $d=16$. Right: percentile relationship between curvature and local OOF MSE at $d=16$. Scale markers use a smaller $n=128$ subset and are not power-matched.

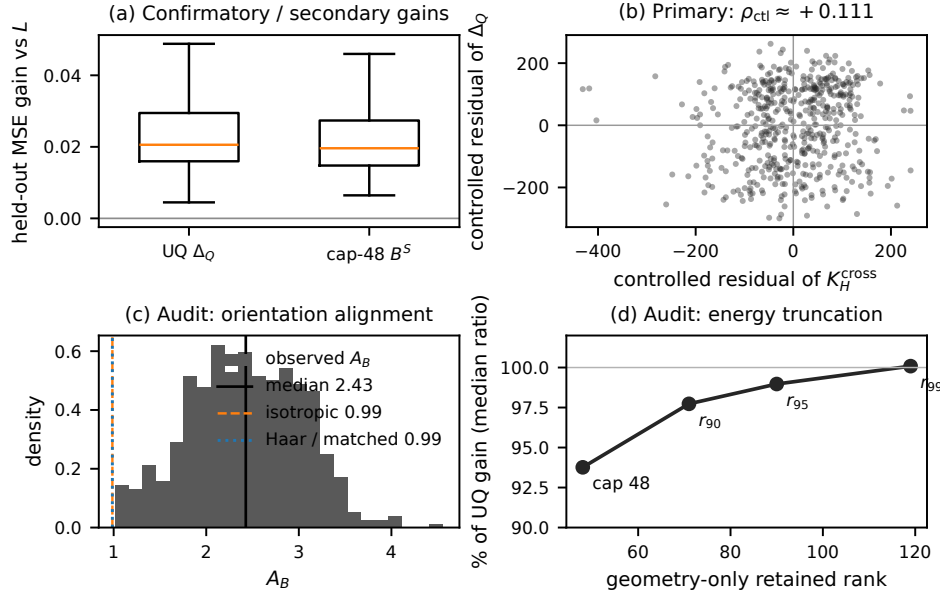


Figure 2: Quadratic decoding (confirmatory a,b; secondary/audit c,d). (a) Held-out MSE gain versus the tangent-linear map L . (b) Controlled rank residuals of K_H^{cross} versus Δ_Q . (c) Observed Hessian–curvature alignment A_B against isotropic and Haar nulls. (d) Median per-anchor fraction of unrestricted quadratic gain retained by geometry-only truncations of B^S .

without labels, recover most of the unrestricted gain as a *median of per-anchor ratios* Δ_{B^S}/Δ_Q : the leading 48 modes retain approximately 94%; 90%/95%/99% energy retain approximately 98%, 99%, and 100% (Fig. 2d).

Hessian–curvature alignment. Normalized alignment $A_B \approx 2.43$ exceeds isotropic, spectrum-preserving Haar, and matched-anchor nulls near 0.99 ($p_{\text{MC}} < 1/2001$; $B=2000$; Fig. 2c). Because the Haar null preserves the singular spectrum of B^S while randomizing orientation, the excess is orientation-sensitive, not a spectral artifact. Foldwise Hessian cosine is approximately 0.92 (all 512 Hessians pass a frozen stability gate). Independent A/B geometry-split alignments both exceed Haar; their Spearman agreement is approximately 0.82. In short, the label Hessian is preferentially oriented toward quadratic coordinate directions along which the representation bends most strongly.

91 **Null calibration and a distinct local-adaptation effect.** On 192 real-design label shuffles of frozen
 92 coordinates, the nested-CV quadratic estimator yields shuffled $\Delta_Q \approx -0.0004$: no false-positive
 93 quadratic gain and satisfactory null calibration. A preregistered fixed-penalty synthetic overfit shuffled
 94 noise because it could not omit the quadratic block; it is not used as evidence against the nested
 95 estimator. Separately, curvature is associated with the relative MSE change from the global probe to
 96 a locally adapted patch probe, $\rho_{\text{ctl}}(K_H, \Delta\text{MSE}_{G \rightarrow P}) \approx +0.153$, but conditioning on Δ_Q raises this
 97 to approximately 0.205. Quadratic chart structure therefore does not mediate that relative adaptation.
 98 Curvature-aligned quadratic decoding and curvature-linked local readout adaptation appear to be
 99 distinguishable phenomena. Patch probes are not claimed to outperform the global probe on average.

100 4 Discussion and limitations

101 Geometry regularizes, rather than algebraically constrains, the full 136-dimensional quadratic space;
 102 predictive concentration onto leading bending modes is stronger than geometric low-rank concentra-
 103 tion. The reported associations are rank- and bandwidth-conditioned, associative rather than causal,
 104 and obtained for one encoder, one imaging dataset, and one photometric observable. Replication
 105 across encoders, datasets, and scientific targets remains future work.

106 **Reproducibility.** Anchors, neighbours, folds, charts, and K_H^{cross} were reused from frozen artifacts
 107 and aligned by sample identity. Unit tests covered quadratic features, fold isolation, and train-only
 108 scaling; a CPU smoke preceded the full 512-anchor run. Primary tables recover prior curvature–probe
 109 correlations to numerical tolerance. Inference uses held-out predictions with 10^4 permutations and
 110 2000 bootstraps, plus real-design label shuffles. Code and frozen artifacts will be released after
 111 review.

112 **Generative AI use.** Generative AI tools assisted with code scaffolding and language editing. All
 113 generated code was covered by targeted unit tests, numerical claims were reproduced from frozen
 114 artifacts, and the authors manually verified the analysis, figures, citations and manuscript.

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